Jorge Manuel Torre

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Education

Virginia Tech Blacksburg, VA
B.S. Computer Science, Overall GPA: 3.37/4.0 May 2026

Experience

Systems Administrator / Infrastructure Engineer January 2019 – December 2023
Independent Systems Administration *Proxmox, Linux, NixOS, Docker, OPNsense*

- Administered and maintained a Linux infrastructure environment spanning Proxmox virtual machines, NixOS systems, and Docker-based services.
- Configured and managed network infrastructure using OPNsense, including firewall rules, port forwarding, DNS, routing, and service accessibility.
- Develop custom scripts, plugins, and server-side software to automate administrative tasks, service management, and system maintenance.
- Managed remote administration through SSH, access controls, service configuration, and secure network practices.
- Deployed and maintain containerized services using Docker, including networking, persistent storage, configuration, and service dependencies.
- Troubleshooted system, application, and network issues using logs, resource utilization, configuration analysis, and command-line tooling.
- Performed software upgrades, configuration changes, backups, and recovery procedures while maintaining service availability.

Projects

Linux Kernel Programming Jan 2026 – Apr 2026
C, x86-64 Assembly, GDB, Linux Kernel, BPF, sched_ext, seccomp

- Implemented custom Linux syscalls with safe user/kernel data transfer; benchmarked syscall and vDSO overhead using `rdtsc`/`rdtscp` and analyzed kernel/user execution paths.
- Instrumented the EEVDF scheduler and implemented a weighted round-robin scheduler using `sched_ext`/BPF; evaluated fairness, latency, and throughput across multicore workloads.
- Built kernel modules for memory profiling and an in-memory pseudo-filesystem, working with page tables, VFS registration, superblocks, inodes, dentries, `seq_file`, and `/proc`; also implemented seccomp-BPF filters and analyzed KPTI page-table isolation in GDB.

Pokémon Black (Nintendo DS) Reverse Engineering Dec 2024 – Jan 2025
ARM Assembly, DeSmuME, No$GBA, PKHeX, Action Replay

- Reverse engineered movement and wild-encounter systems using ARM assembly, memory tracing, and known-good Action Replay patches; identified live instructions and encounter tables and verified behavior through memory injection.
- Reconstructed the Gen 5 in-memory Pokémon data layout by correlating RAM dumps with PKHeX, identifying encrypted 220-byte party structures containing species, IVs/EVs, nickname, OT, and nature.

Computer Systems Dec 2024 – Jan 2025
C, POSIX APIs, pthreads, sockets, HTTP/1.1, JWT

- Built a Unix shell supporting pipelines, I/O redirection, background jobs, process groups, signals, command substitution, and control-flow constructs; implemented a fork/join thread pool with futures, work stealing, and per-worker deques.
- Implemented a concurrent HTTP/1.1 server with persistent connections, MP4 streaming, JWT authentication, and path-traversal protection; built a dynamic memory allocator with heap consistency checking.

Leadership & Community

Team Lead, Virginia Tech CS Projects Jan 2025 – Aug 2025

- Led 3–5 person teams through requirements gathering, schema design, SQL development, sprint planning, code reviews, integration, and final project delivery.

Garage32 – Community Project Mar 2020 – Dec 2021

- Led development and distribution of a Minecraft graphical asset pack reaching **600+ downloads**; promotional content generated **30K+ likes** on a single TikTok post.

Skills

Languages: C, C++, Python, Java, SQL, JavaScript, Bash, ARM/x86-64/RISC-V Assembly
Systems: Linux Kernel, POSIX, Concurrency, GDB, Ghidra, BPF/sched_ext, seccomp, Reverse Engineering
Infrastructure: Linux, Docker, Proxmox, NixOS, OPNsense, PostgreSQL, Git, CI/CD